

Robust Regression

Dr. Kiah Wah Ong

Introduction: Robustness

Suppose we have the following data $x_1, \dots, x_n$ and consider the average

$$\begin{aligned}\bar{x} &= \frac{1}{n} \sum_{i=1}^n x_i \\ &= \frac{1}{n} \left[\sum_{i=1}^{n-1} x_i + x_n \right] \\ &= \frac{n-1}{n} \bar{x}_{n-1} + \frac{1}{n} x_n\end{aligned}$$

Notice that if the value of x_n is large enough, then the average, $\bar{x}$ can be made as large as possible regardless of the other $n-1$ values of x_i .

Because of this, we say that the mean (average) is not a robust measure of central tendency.

Least Squares Method

Recall that in our simple linear regression model

$$y = \beta_0 + \beta_1 x + \epsilon$$

the parameters β_0 and β_1 are unknown and must be estimated using sample data

$$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n).$$

The least squares method is used to estimate β_0 and β_1 . To do this, from

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

we write

$$\epsilon_i = y_i - (\beta_0 + \beta_1 x_i)$$

and we want to find $\hat{\beta}_1$ and $\hat{\beta}_2$ that minimize the residual sum of squares.

$$S(\beta_0, \beta_1) = \sum_{i=1}^n \epsilon_i^2 = \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_i))^2$$

Least Squares Method

Because of squares in

$$S(\beta_0, \beta_1) = \sum_{i=1}^n \epsilon_i^2 = \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_i))^2$$

outliers are heavily weighted. Bad data points can throw off the regression.

M-Estimation

M-estimation is a generalization of the ordinary least squares. (M for Maximum)

Instead of squaring the residuals, we will define a general function of the residuals, $H(\epsilon_i)$, and then minimize

$$S = \sum H(\epsilon_i).$$

M-Estimation

We want the function H has the following properties:

- ▶ H is **non-negative**, i.e. $H(\epsilon_i) \geq 0$
- ▶ $H(0) = 0$
- ▶ H is **symmetric**, i.e. $H(-\epsilon_i) = H(\epsilon_i)$
- ▶ H is **monotonic**, i.e.
if $|\epsilon_i| > |\epsilon_j|$, then $H(\epsilon_i) > H(\epsilon_j)$
- ▶ H has **continuous derivative** with respect to the coefficients.
This allows us in finding the minimum more effectively.

Notice that $H(\epsilon_i) = \epsilon_i^2$ has all the above properties.

M-Estimation

For the ordinary least squares method, we let $S = \sum_{i=1}^n \epsilon_i^2$ and then take the derivative with respect to a parameter and set that equal to zero:

$$\frac{\partial S}{\partial \beta_j} = 0 \quad \text{gives} \quad \sum_{i=1}^n \epsilon_i x_{ji} = 0$$

For M-estimator, with $S = \sum H(\epsilon_i)$, we have

$$\frac{\partial S}{\partial \beta_j} = 0 \quad \text{gives} \quad \sum_{i=1}^n \frac{\partial H}{\partial \epsilon_i} x_{ji} = 0$$

M-Estimation

Let us define the weight as

$$w_i = \frac{1}{\epsilon_i} \frac{\partial H}{\partial \epsilon_i}$$

Hence, the previous expression

$$\sum_{i=1}^n \frac{\partial H}{\partial \epsilon_i} x_{ji} = 0$$

can be written as

$$\sum_{i=1}^n \frac{\partial H}{\partial \epsilon_i} x_{ji} = 0 = \sum_{i=1}^n w_i \epsilon_i x_{ji}$$

that is, we have the set up of a weighted linear regression. This gives us a scheme that we can use to do M-estimation.

M-Estimation

In order to perform the M-estimation from

$$w_i = \frac{1}{\epsilon_i} \frac{\partial H}{\partial \epsilon_i}, \quad \sum_{i=1}^n w_i \epsilon_i x_{ji} = 0$$

we will:

- ▶ guess the weight, w_i
- ▶ calculate the residuals, ϵ_i
- ▶ use the previous residuals to calculate new weight
- ▶ we repeat these process until convergence

The process described above are called iteratively reweighted least squares.

M-Estimation

What we want to do now is to find a good function H that has the following properties that can give rise to good result.

- ▶ H is **non-negative**, i.e. $H(\epsilon_i) \geq 0$
- ▶ $H(0) = 0$
- ▶ H is **symmetric**, i.e. $H(-\epsilon_i) = H(\epsilon_i)$
- ▶ H is **monotonic**, i.e.
if $|\epsilon_i| > |\epsilon_j|$, then $H(\epsilon_i) > H(\epsilon_j)$
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Huber M-Estimator

One of the commonly used M-estimator is the Huber M-estimator given below:

$$H(\epsilon) = \begin{cases} \epsilon^2/2, & \text{for } |\epsilon| \leq k \\ k|\epsilon| - k^2/2 & \text{for } |\epsilon| > k \end{cases}$$

Notice that when within a certain value of k , the function H behaves like the ordinary least square estimator, however, when we went beyond the value of k , then we will switch to using the absolute value of the residue (hence avoiding amplifying the effect of outliers.)

The value k for the Huber estimator is called a tuning constant. The tuning constant is picked to give a high efficiency when the data is normal.

Huber chooses the value of $k = 1.345\sigma$ that produces a 95% efficiency when the errors are normal, and still offers protection against outliers.

Bisquare M-Estimator

Another commonly used M-estimator is called Bisquare M-Estimator.

$$H(\epsilon) = \begin{cases} k^2/6 (1 - [1 - (\epsilon/k)^2]^3) & \text{for } |\epsilon| \leq k \\ k^2/6 & \text{for } |\epsilon| > k \end{cases}$$

The bisquare M-estimator in a sense is more robust than the Huber M-estimator. This is because when $|\epsilon| > k = 4.685\sigma$, the weighting become constant.

M-estimator Weights

Another way of looking at these two M-estimators is by looking at the weight, w_i in:

$$w_i = \frac{1}{\epsilon_i} \frac{\partial H}{\partial \epsilon_i}, \quad \sum_{i=1}^n w_i \epsilon_i x_{ji} = 0$$

For Huber M-estimator, we have

$$w(\epsilon) = \begin{cases} 1 & \text{for } |\epsilon| \leq k \\ k/|\epsilon| & \text{for } |\epsilon| > k \end{cases}$$

While, for bisquare M-estimator, we have

$$w(\epsilon) = \begin{cases} \left[1 - \left(\frac{\epsilon}{k}\right)^2\right]^2 & \text{for } |\epsilon| \leq k \\ 0 & \text{for } |\epsilon| > k \end{cases}$$

M-estimator Weights

For the Huber M-estimator,

$$w(\epsilon) = \begin{cases} 1 & \text{for } |\epsilon| \leq k \\ k/|\epsilon| & \text{for } |\epsilon| > k \end{cases}$$

we see that the weight started at one and then weighting less after some point.

For bisquare M-estimator,

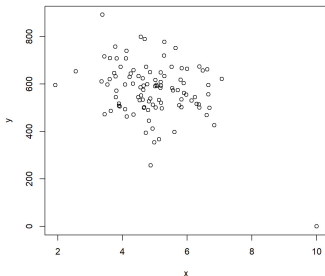
$$w(\epsilon) = \begin{cases} \left[1 - \left(\frac{\epsilon}{k}\right)^2\right]^2 & \text{for } |\epsilon| \leq k \\ 0 & \text{for } |\epsilon| > k \end{cases}$$

we see that the weight immediately started to drop off from one (when $\epsilon = 0$) to zero after $|\epsilon| > k$. Hence extreme outliers will not be weighted at those level.

Robust Regression in R

We now look at how to perform robust regression in *R*: Let us import the simulated data from data set `RRobust1.csv` and plot the data.

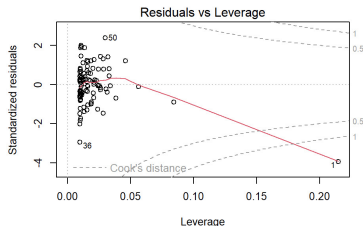
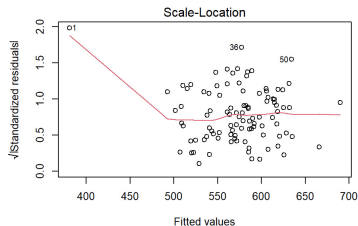
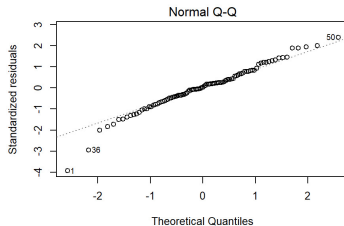
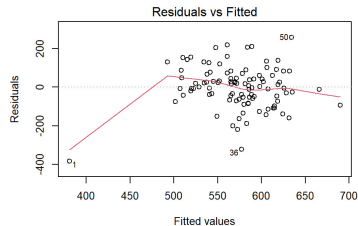
```
Robustregression<-read.csv("RRobust1.CSV", header=TRUE, sep=",")  
x<-Robustregression$x  
y<-Robustregression$y  
plot(x,y)
```



We can see clearly that the data point $(10, -1)$ is an outlier.

Robust Regression in R

The standard diagnostic plots also make a clear point that $(10, -1)$ is an outlier.

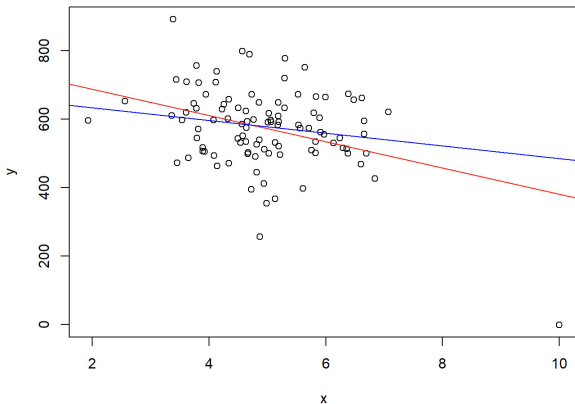


Robust Regression in R

We now make two regression models. In `model1`, an OLS regression is performed, and the regression line is plot in red. In `model2`, an OLS regression is performed with the outlier removed. This regression line is shown in blue.

```
#With outlier
model1=lm(y~x)
abline(model1, col="red")

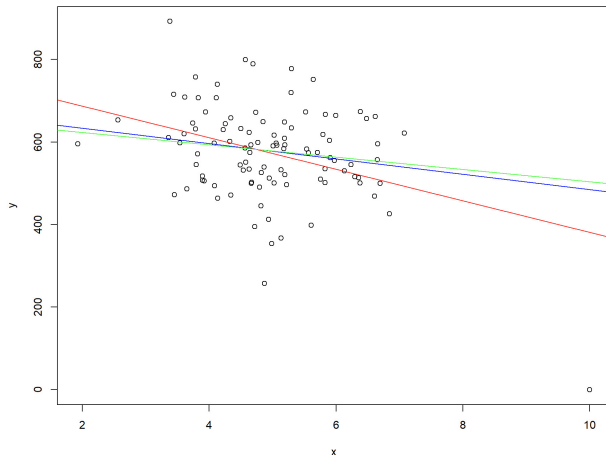
#Removing outlier
df1<-Robustregression
df2<-df1[-1,]
model2=lm(y~x,data=df2)
abline(model2, col="blue")
```



Robust Regression in R

Now, model3 is set up as a robust regression model. The outlier is now weighted down by using the bisquare method.

```
library(MASS)
#Not removing outlier but use robust regression
model3=rlm(y~x, data=df1, psi=psi.bisquare)
abline(model3, col="green")
```



Robust Regression in R

Using robust regression technique, we see that there is no need to pretreat the outliers.

